

অবশ্যই। নিচে শুধু Basic Definition + ছোট Graphical Diagram দিলাম—exam-এর আগে দ্রুত revision করার জন্য। বড় explanation বাদ দিলাম।

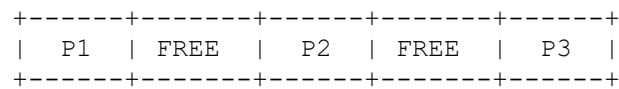
OS Memory Management — Basic Definitions with Diagram

1. Fragmentation

Definition:

Fragmentation is the wastage of memory space caused by memory being divided into small or unusable portions.

বাংলা: Memory-এর কিছু জায়গা ব্যবহার না হওয়াকে Fragmentation বলে।

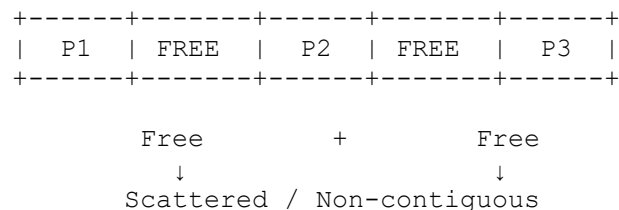


2. External Fragmentation

Definition:

External fragmentation occurs when sufficient total free memory exists, but it is scattered into non-contiguous blocks.

বাংলা: মোট free memory যথেষ্ট থাকলেও contiguous না থাকলে External Fragmentation হয়।

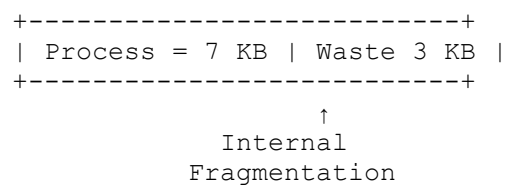


3. Internal Fragmentation

Definition:

Internal fragmentation occurs when unused memory exists inside an allocated memory block.

বাংলা: Allocated block-এর ভিতরে unused space থাকলে Internal Fragmentation হয়।



4. Compaction

Definition:

Compaction is the process of moving processes together to combine scattered free spaces into one large contiguous block.

বাংলা: ছড়িয়ে থাকা free space-গুলোকে এক জায়গায় এনে বড় একটি block তৈরি করাকে Compaction বলে।

Before:

```
+---+---+---+---+---+
| P1 | FREE | P2 | FREE | P3 |
+---+---+---+---+---+
```

After:

```
+---+---+---+---+---+
| P1 | P2 | P3 |   FREE   |
+---+---+---+---+---+
```

Dynamic Memory Allocation

5. First-Fit

Definition:

First-fit allocates the **first hole that is large enough** for the process.

বাংলা: শুরু থেকে search করে প্রথম suitable hole-এ process বসায়।

```
H1      H2      H3
100      500      300 KB
          ↑
        First suitable
```

6. Next-Fit

Definition:

Next-fit starts searching from the **location where the previous search ended**.

বাংলা: আগের allocation যেখানে শেষ হয়েছিল, সেখান থেকে search শুরু করে।

```
H1 → H2 → H3 → H4 → H5
          ↑
        Last position
          ↓
        Search →
```

7. Best-Fit

Definition:

Best-fit allocates the **smallest hole that is large enough**.

বাংলা: Suitable hole-গুলোর মধ্যে সবচেয়ে ছোট hole নির্বাচন করে।

```

100 KB    500 KB    300 KB    600 KB
           ↑
        Best Fit
  
```

8. Worst-Fit

Definition:

Worst-fit allocates the **largest available hole**.

বাংলা: সবচেয়ে বড় available hole-এ process বসায়।

```

100 KB    500 KB    300 KB    600 KB
                               ↑
                          Worst Fit
  
```

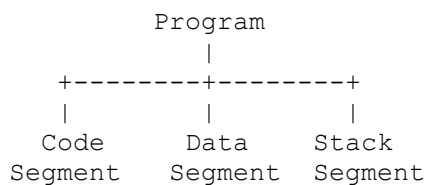
Segmentation & Paging

9. Segmentation

Definition:

Segmentation divides a program into **variable-sized logical segments**.

বাংলা: Program-কে Code, Data, Stack ইত্যাদি logical এবং variable-size অংশে ভাগ করাকে Segmentation বলে।

**Address:**

< Segment Number, Offset >

10. Segment Table

Definition:

A segment table stores the **base address and limit** of each segment.

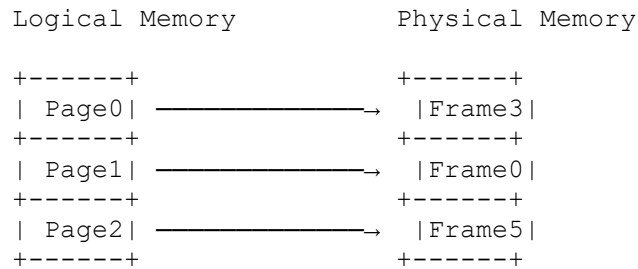
Segment	Base	Limit
0	1000	400
1	5000	600
2	8000	300

11. Paging

Definition:

Paging divides logical memory into fixed-size **pages** and physical memory into fixed-size **frames**.

বাংলা: Logical memory-কে Page এবং Physical memory-কে Frame-এ fixed size-এ ভাগ করাকে Paging বলে।



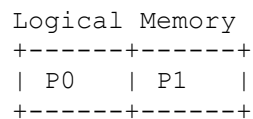
Important:

Page Size = Frame Size

12. Page

Definition:

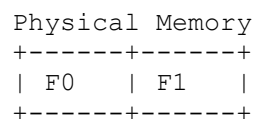
A page is a fixed-size block of **logical/virtual memory**.



13. Frame

Definition:

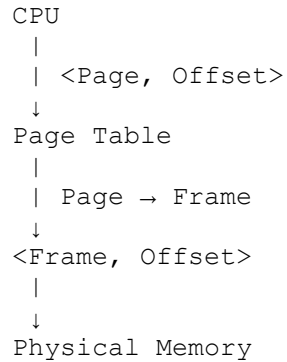
A frame is a fixed-size block of **physical memory (RAM)**.



14. Address Translation in Paging

Definition:

Address translation converts a logical address into a physical address using the page table.



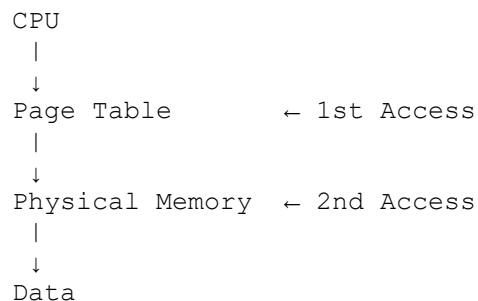
Key:

Page Number $\rightarrow$ Frame Number
Offset $\rightarrow$ Same

15. Two-Memory-Access Problem

Definition:

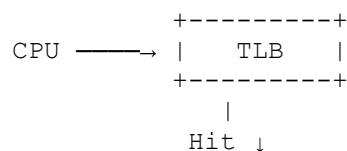
In paging, two memory accesses may be required: one for the page table and another for the actual data.

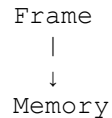


16. TLB

Definition:

TLB (Translation Lookaside Buffer) is a small, fast cache that stores recently used page-to-frame mappings.





17. TLB Hit

Definition:

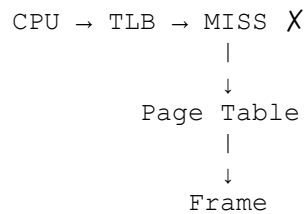
A TLB hit occurs when the required page number is found in the TLB.

CPU → TLB → HIT ✓ → Frame → Memory

18. TLB Miss

Definition:

A TLB miss occurs when the required page number is not found in the TLB.



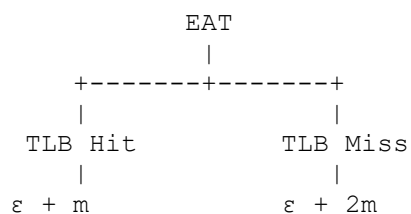
19. Effective Access Time (EAT)

Definition:

EAT is the average memory access time considering both TLB hits and misses.

Formula:

$$[\text{EAT} = \alpha(\epsilon + m) + (1 - \alpha)(\epsilon + 2m)]$$

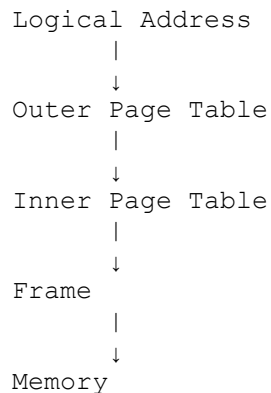


Advanced Paging

20. Hierarchical Paging

Definition:

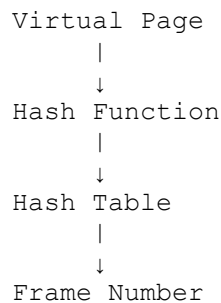
Hierarchical paging divides a large page table into multiple levels.



21. Hashed Page Table

Definition:

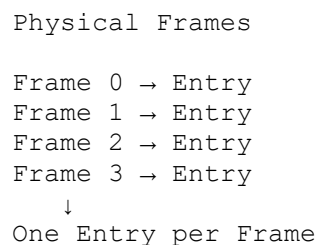
A hashed page table uses a hash function on the virtual page number to find the corresponding page-table entry.



22. Inverted Page Table

Definition:

An inverted page table has **one entry for each physical frame** rather than for each virtual page.

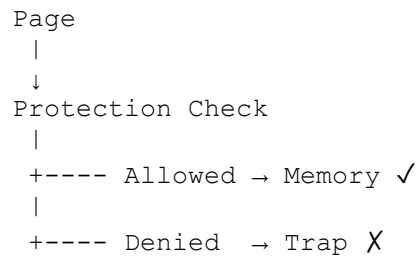


Sharing & Protection

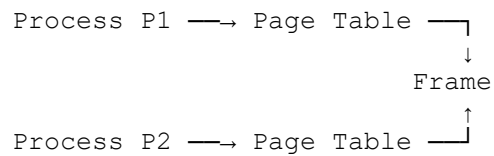
23. Memory Protection

Definition:

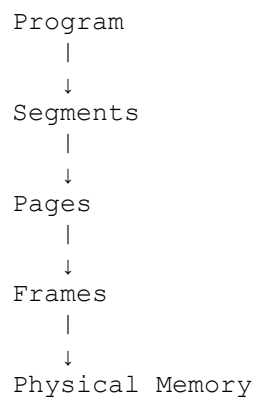
Memory protection prevents unauthorized access to memory by using access-control bits such as Read, Write, and Execute.

**24. Shared Pages****Definition:**

Shared pages allow multiple processes to map their logical pages to the same physical frame.

**25. Combined Segmentation and Paging****Definition:**

Combined segmentation and paging divides a program into logical segments and then divides each segment into fixed-size pages.

**Address:**

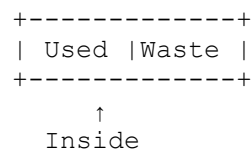
< Segment, Page, Offset >

★ Important Differences

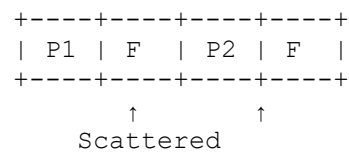
26. Internal vs External Fragmentation

Feature	Internal	External
Location	Inside allocated block	Outside allocated blocks
Block	Allocated block is larger	Free blocks are scattered
Main problem	Wasted space inside	Non-contiguous free space
Example	7 KB used in 10 KB block	10 KB + 8 KB free but need 20 KB

Internal:



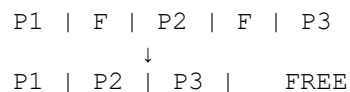
External:



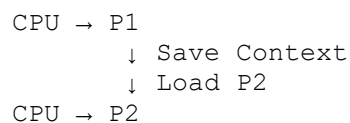
27. Compaction vs Context Switching

Feature	Compaction	Context Switching
Area	Memory Management	Process Management
Purpose	Reduce external fragmentation	Switch CPU between processes
What changes?	Process memory locations	CPU execution state
Main operation	Move/relocate processes	Save & restore context
RAM movement	Yes	Normally no

Compaction:



Context Switching:



28. Segmentation vs Paging

Feature	Segmentation	Paging
Unit	Segment	Page
Size	Variable	Fixed
Based on	Logical view	Physical memory management
Address	Segment + Offset	Page + Offset
Fragmentation	External	Internal
Visibility	Programmer-visible concept	Generally invisible
Mapping	Segment → Memory	Page → Frame

Segmentation:

```
<Segment, Offset>
  ↓
Segment Table
  ↓
Physical Memory
```

Paging:

```
<Page, Offset>
  ↓
Page Table
  ↓
<Frame, Offset>
  ↓
Physical Memory
```

29. SCAN / Elevator Algorithm

Definition:

SCAN is a disk scheduling algorithm in which the disk head moves in one direction, services requests, reaches the end, and then reverses direction.

```
0 ----- 199
    20    40    60    80   100
          ↑
        Head

40 → 60 → 80 → 100 → 199
                      ↓
199 → 100 → 80 → 60 → 40 → 20
```

Memory Trick:

SCAN = Go → Serve → End → Reverse → Serve